



12-B Status from UGC

**BCA III Semester
Operating System
Course Code: BCAC0020**

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OS

Introduction of Operating System

- An Operating System (OS) is a collection of program which provides an interface between a computer user and computer hardware.
- An operating system is a software which performs all the basic tasks like file management, memory management, process management, handling input and output, and controlling peripheral devices such as disk drives and printers.
- A Collection program that acts as an intermediary between a user of a computer and the computer hardware.
- It is responsible for the management and coordination of activities and the sharing of the resources of the computer.

Computer System Components

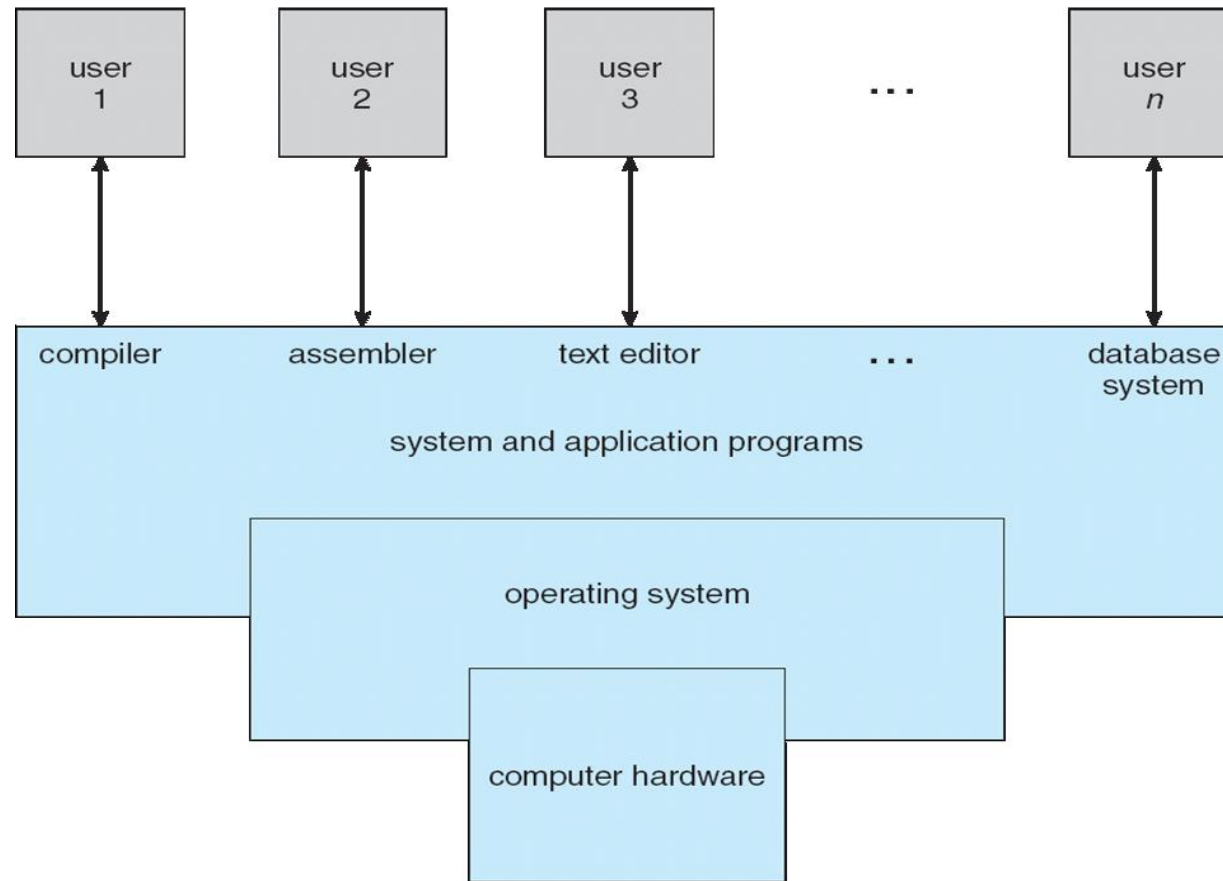
Hardware Provides basic computing resources (CPU, memory, I/O devices).

Operating System Controls and coordinates the use of hardware among application programs.

Application Programs Solve computing problems of users (compilers, database systems, video games, business programs such as banking software).

Users People, machines, other computers

Four Components of a Computer System



Function of Operating System

1. Process management

A program in its execution state is known as **process**.

A process needs certain resources including CPU time, memory, files and I/O devices to accomplish its task.

These resources are either given to the process when it is created or allocated to it while it is running.

A program is a passive *entity* such as contents of a file stored on the disk whereas a process is an active *entity*

The operating system is responsible for the following activities in process management:

- Creating and deleting both user and system processes
- Suspending and resuming processes
- Providing mechanisms for process synchronization
- Providing mechanisms for process communication
- Providing mechanisms for deadlock handling.

2. Memory management

Main memory is a collection of quickly accessible data shared by the CPU and I/O devices.

The central processor reads instructions from main memory (during instruction-fetch cycle) and both reads and writes data from main memory (during data-fetch cycle)

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The operating system is responsible for the following activities in memory management:

- Keeping track of which parts of memory are currently being used and by whom
- Deciding which processes and data to move into and out of memory
- Allocating and deallocating memory space as needed.

3. File-System Management

The operating system is responsible for the following activities with file management:

- Creating and deleting files
- Creating and deleting directories to organize files
- Supporting primitives for manipulating files and directories
- Backing up files on stable (nonvolatile) storage media.

4. Secondary storage Management The operating system is responsible for the following activities with disk management:

- Free-space management
- Storage allocation
- Disk scheduling

5. I/O system Management

The operating system is responsible for the following activities with I/O subsystem:

A memory-management component that includes buffering, caching, and spooling

A general device-driver interface

Drivers for specific hardware devices

6. Protection and Security

A mechanisms ensure that files, memory segments, CPU, and other resources can be operated on by only those processes that have been allowed proper authorization from the operating system.

For example, memory-addressing hardware ensures that a process can execute only within its own address space.

Protection is a mechanism for controlling the access of processes or users to the resources defined by a computer system.

Classification of Operating Systems

An operating system is a well-organized collection of programs that manages the computer hardware. It is a type of system software that is responsible for the smooth functioning of the computer system.

- Batch Operating System
- Real Time Operating System
- Multiprogramming Operating System
- Multitasking Operating System
- Multiprocessing Operating System
- Time-Sharing Operating System

Types of Operating Systems (OS)

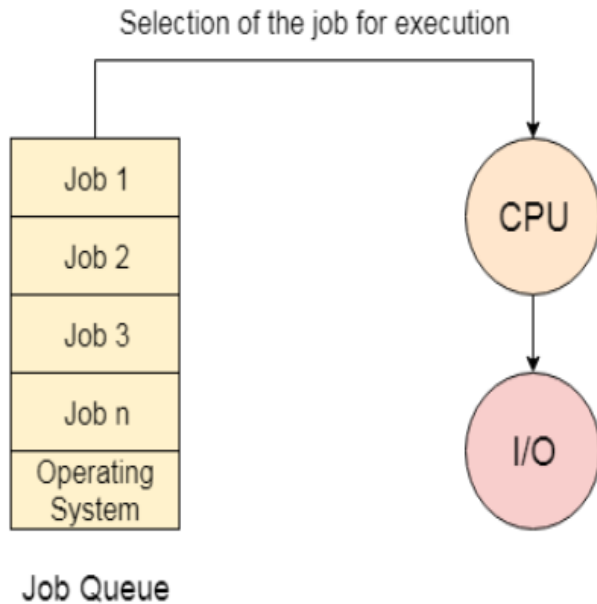
Batch Operating System

In the 1970s, Batch processing was very popular. In this technique, similar types of jobs were batched together and executed in time. People were used to having a single computer which was called a mainframe.

In Batch operating system, access is given to more than one person; they submit their respective jobs to the system for the execution.

The system put all of the jobs in a queue on the basis of first come first serve and then executes the jobs one by one. The users collect their respective output when all the jobs get executed.

For Example:



Interactive,

There are five jobs J1, J2, J3, J4, and J5, present in the batch. If the execution time of J1 is very high, then the other four jobs will never be executed, or they will have to wait for a very long time. Hence the other processes get starved.

Interactive

- An interactive operating system gives permission to the user to interact directly with the computer. In an Interactive operating system, the user enters some command into the system and the work of the system is to execute it.
- An interactive operative system is an operating system that allows the execution of interactive programs. All PC operating systems are interactive operating systems only.

Real Time Operating System

- A real time system has well defined fixed time constraints, processing must be done within defined constraints or system will get failed.
- System that controls scientific system, experimenting medical system, industrial control system and certain display systems are real time system.
- They are also applicable to automobile engine fuel system, home appliance controller and weapon systems.
- There are two types of real system:

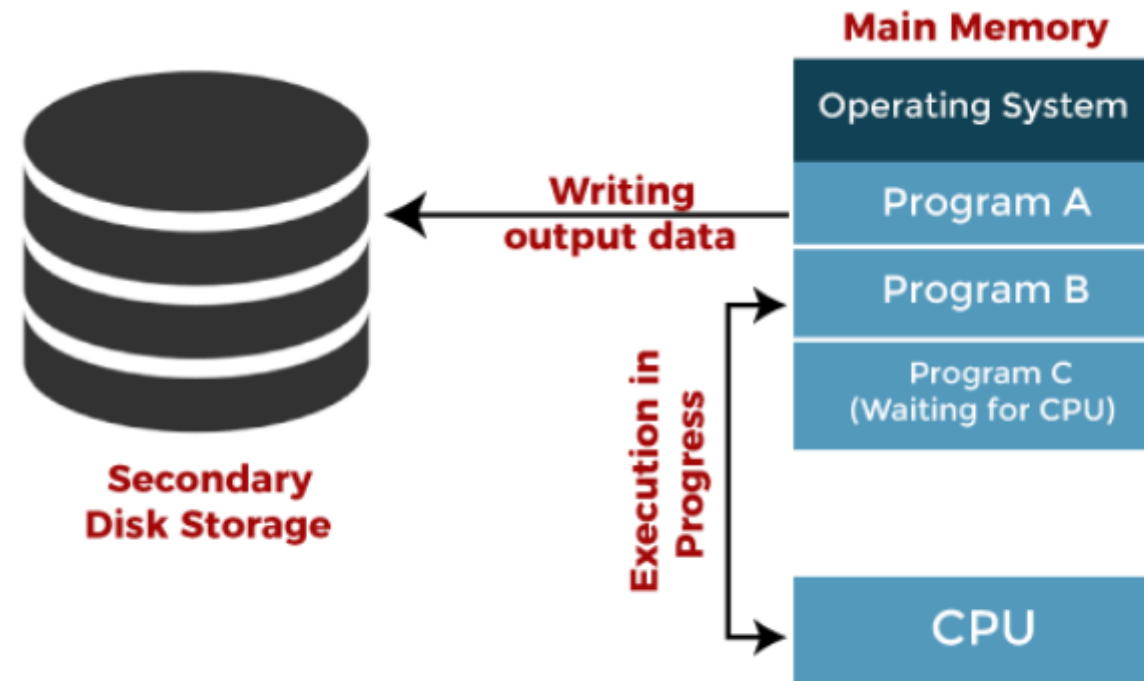
i) Hard real time system This system guarantees that critical tasks be completed on time. For this all the delays in the system should be bounded, from the retrieval of stored data to the time it takes operating system to finish any request made to it.

ii) Soft real time system This is less restrictive type of system defined as **not hard real-time**, simply providing that a critical real-time task will receive priority over other tasks and that it will retain the priority until it completes.

Multiprogramming Operating System

Multiprogramming is an extension to batch processing where the CPU is always kept busy. Each process needs two types of system time: CPU time and IO time.

In a multiprogramming environment, when a process does its I/O, The CPU can start the execution of other processes. Therefore, multiprogramming improves the efficiency of the system.



Jobs in multiprogramming system

Advantages of Multiprogramming OS

Throughout the system, it increased as the CPU always had one program to execute.

Response time can also be reduced.

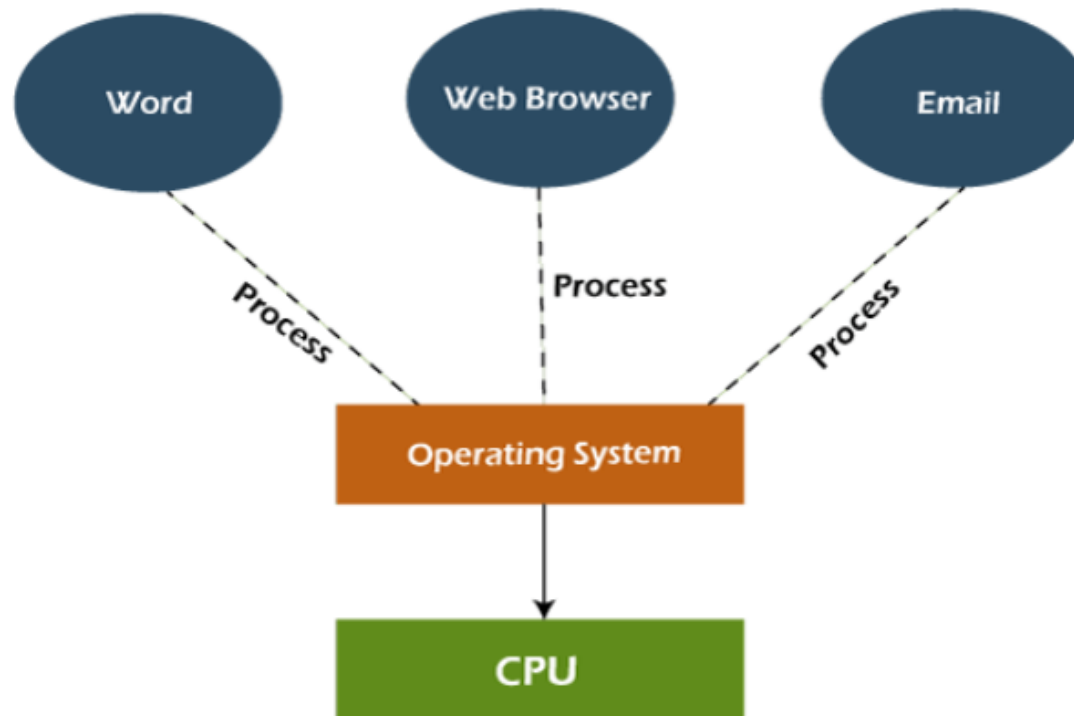
Disadvantages of Multiprogramming OS

Multiprogramming systems provide an environment in which various systems resources are used efficiently, but they do not provide any user interaction with the computer system.

Multitasking Operating System

The multitasking operating system is a logical extension of a multiprogramming system that enables **multiple** programs simultaneously. It allows a user to perform more than one computer task at the same time.

Multitasking Operating System



Advantages of Multitasking operating system

This operating system is more suited to supporting multiple users simultaneously.

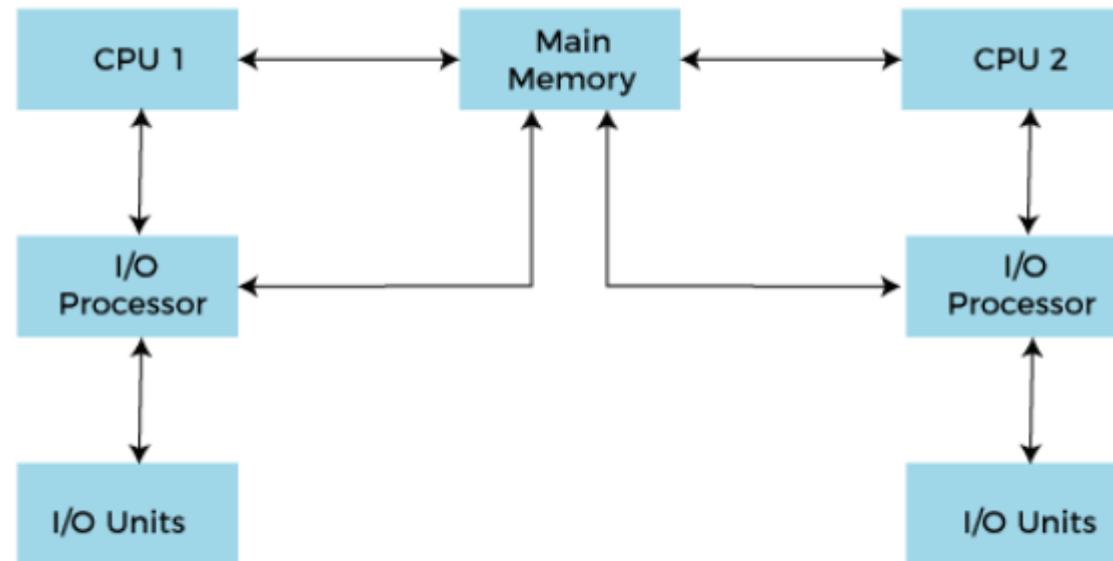
The multitasking operating systems have well-defined memory management.

Disadvantages of Multitasking operating system

The multiple processors are busier at the same time to complete any task in a multitasking environment, so the CPU generates more heat.

Multiprocessing Operating System

In Multiprocessing, Parallel computing is achieved. There are more than one processors present in the system which can execute more than one process at the same time. This will increase the throughput of the system.



Working of Multiprocessor System

Advantages of Multiprocessing operating system:

Increased reliability: Due to the multiprocessing system, processing tasks can be distributed among several processors. This increases reliability as if one processor fails, the task can be given to another processor for completion.

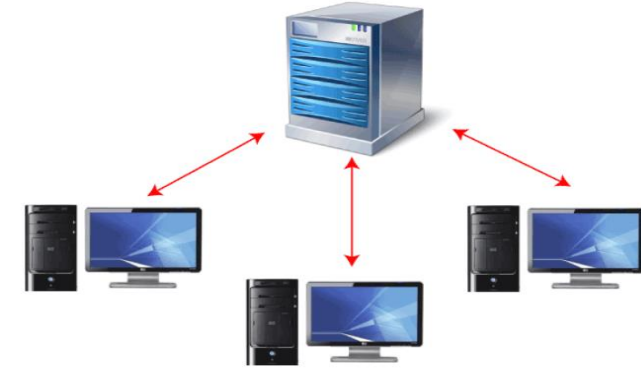
Increased throughput: As several processors increase, more work can be done in less.

Disadvantages of Multiprocessing operating System

Multiprocessing operating system is more complex and sophisticated as it takes care of multiple CPUs simultaneously.

Multi-User Operating System

A multi-user operating system is an operating system that permits several users to access a single system running to a single operating system. Users will usually sit at computers connected to the system via a network and other system machines like printers.



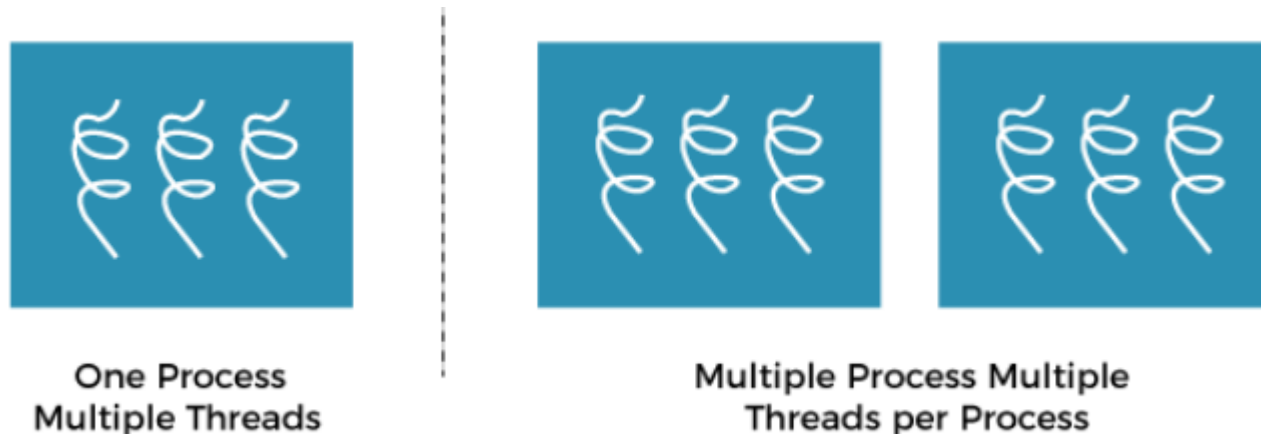
Multiple numbers of users can access different resources of a computer at the same time. The access is provided using a network that consists of various personal computers attached to a mainframe computer system. A multi-user operating system allows the permission of multiple users for accessing a single machine at a time. The various personal computers can send and receive information to the mainframe computer system. Thus, the mainframe computer acts as the server and other personal computers act as clients for that server.

Multithreaded Systems

A thread is a path of execution within a process or Thread is a sequence flow of task within a process.

Multithreading allows the application to divide its task into individual threads. In multi-threads, the same process or task can be done by the number of threads, or we can say that there is more than one thread to perform the task in multithreading. With the use of multithreading, multitasking can be achieved.

The main drawback of single threading systems is that only one task can be performed at a time, so to overcome the drawback of this single threading, there is multithreading that allows multiple tasks to be performed



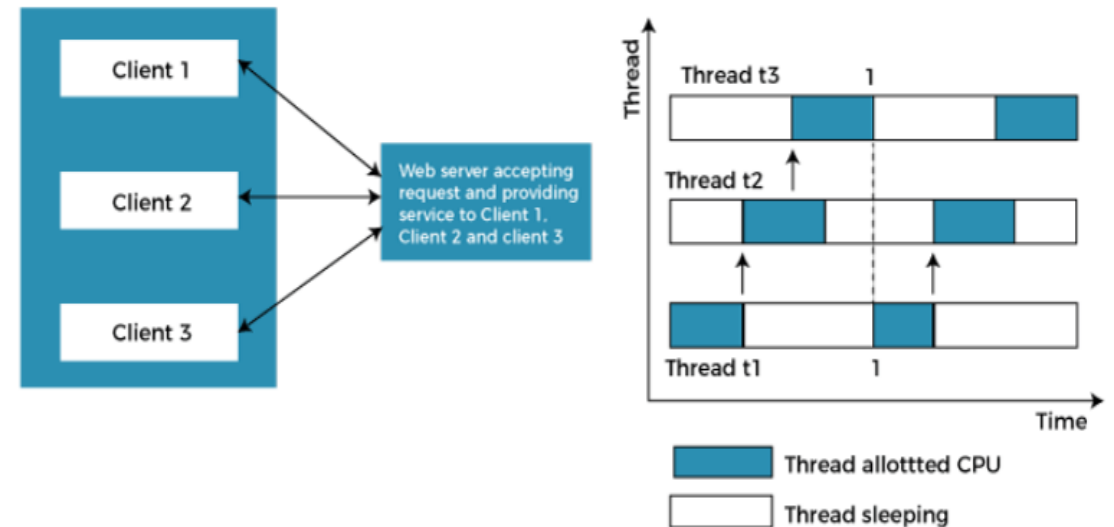
Multithreaded Systems

For example:

In this example, client1, client2, and client3 are accessing the web server without any waiting. In multithreading, several tasks can run at the same time.

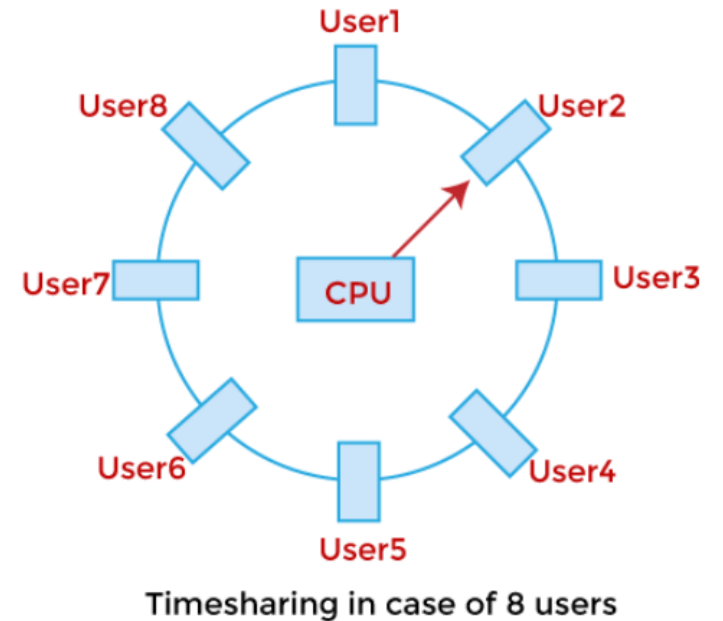
In an operating system, threads are divided into the user-level thread and the Kernel-level thread. User-level threads handled independent form above the kernel and thereby managed without any kernel support.

On the opposite hand, the operating system directly manages the kernel-level threads. Nevertheless, there must be a form of relationship between user-level and kernel-level threads.



Time-Sharing Operating System

In the Time Sharing operating system, computer resources are allocated in a time-dependent fashion to several programs simultaneously. Thus it helps to provide a large number of user's direct access to the main computer. It is a logical extension of multiprogramming. In time-sharing, the CPU is switched among multiple programs given by different users on a scheduled basis.



A time-sharing operating system allows many users to be served simultaneously, so sophisticated CPU scheduling schemes and Input/output management are required.

Time-sharing operating systems are very difficult and expensive to build.

Advantages of Time Sharing Operating System

The time-sharing operating system provides effective utilization and sharing of resources.

This system reduces CPU idle and response time.

Disadvantages of Time Sharing Operating System

Data transmission rates are very high in comparison to other methods.

Security and integrity of user programs loaded in memory and data need to be maintained as many users access the system at the same time.

Operating System Structure

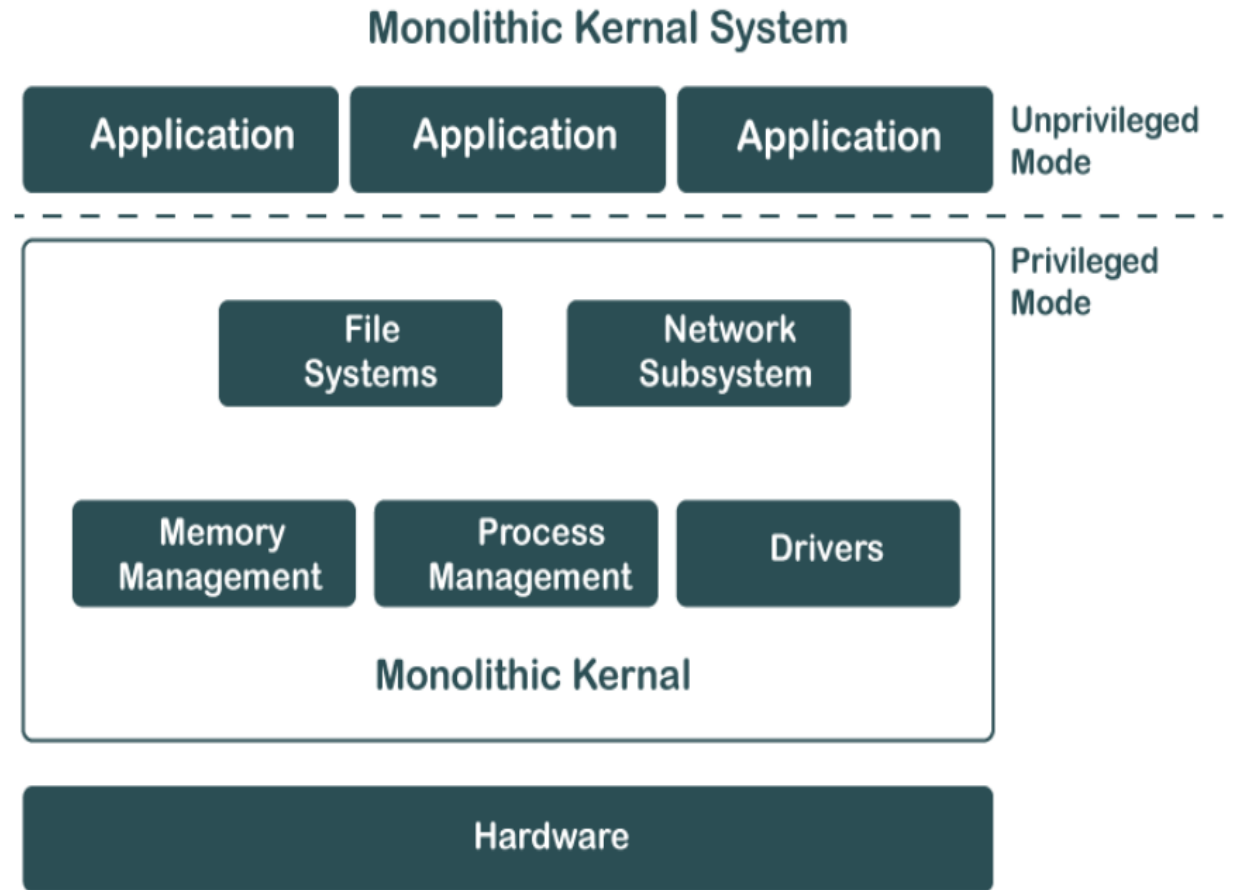
An operating system is a design that enables user application programs to communicate with the hardware of the machine. The operating system should be built with the most care because it is such a complicated structure and should be simple to use and modify. Partially developing the operating system is a simple approach to accomplish this. Each of these components needs to have distinct inputs, outputs, and functionalities.

- **Monolithic Structure**
- **Micro-Kernel Structure**

Monolithic Structure-The **monolithic** operating system controls all aspects of the operating system's operation, including file management, memory management, device management, and operational operations.

The core of an operating system for computers is called the kernel (OS). All other System components are provided with fundamental services by the kernel. The operating system and the hardware use it as their main interface. **When an operating system is built into a single piece of hardware, such as a keyboard or mouse, the kernel can directly access all of its resources.**

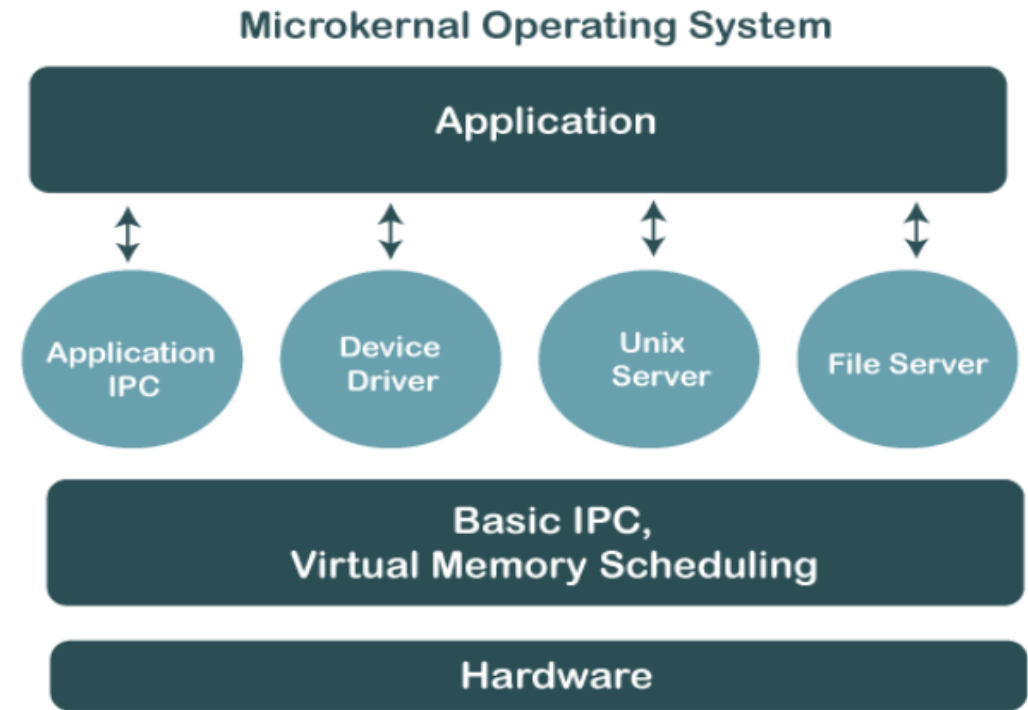
The following diagram represents the monolithic structure:



MICRO-KERNEL STRUCTURE

The operating system is created using a micro-kernel framework that strips the kernel of any unnecessary parts. Systems and user applications are used to implement these optional kernel components. Each Micro-Kernel is created separately and is kept apart from the others. As a result, the system is now more trustworthy and secure. If one Micro-Kernel malfunctions, the remaining operating system is unaffected and continues to function normally. The image below shows Micro-Kernel Operating System Structure:

The image below shows Micro-Kernel Operating System Structure:



SCHEDULERS

A process migrates between the various scheduling queues throughout its life-time purposes. The OS must select for scheduling processes from these queues in some fashion. This selection process is carried out by the appropriate scheduler. In a batch system, more processes are submitted and then executed immediately. So these processes are spooled to a mass storage device like disk, where they are kept for later execution.

Types of schedulers: There are 3 types of schedulers mainly used:

1. Long term scheduler:-Long term scheduler selects process from the disk & loads them into memory for execution. It controls the degree of multi-programming i.e. no. of processes in memory. It executes less frequently than other schedulers. If the degree of multiprogramming is stable than the average rate of process creation is equal to the average departure rate of processes leaving the system. So, the long term scheduler is needed to be invoked only when a process leaves the system. Due to longer intervals between executions it can afford to take more time to decide which process should be selected for execution. Most processes in the CPU are either I/O bound or CPU bound.

2. Short - term scheduler: The short term scheduler selects among the process that are ready to execute & allocates the CPU to one of them. The primary distinction between these two schedulers is the frequency of their execution. The short-term scheduler must select a new process for the CPU quite frequently. It must execute at least one in 100ms. Due to the short duration of time between executions, it must be very fast.

3. Medium - term scheduler: some operating systems introduce an additional intermediate level of scheduling known as medium - term scheduler. The main idea behind this scheduler is that sometimes it is advantageous to remove processes from memory & thus reduce the degree of multiprogramming. At some later time, the process can be reintroduced into memory & its execution can be continued from where it had left off. This is called as swapping. The process is swapped out & swapped in later by medium term scheduler. Swapping is necessary to improve the process miss or due to some change in memory requirements, the available memory limit is exceeded which requires some memory to be freed up.

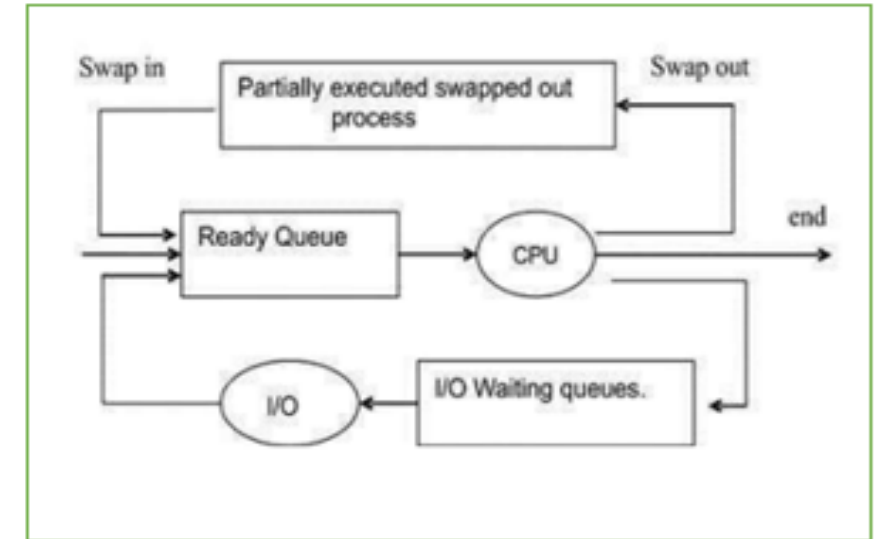
Process Control Block

Each process is represented in the OS by a process control block. It is also by a process control block. It is also known as task control block.

A process control block contains many pieces of information associated with a specific process. It includes the following information's.

pointer	process state
process number	
program counter	
registers	
memory limits	
list of open files	
⋮	

- **Process state:** The state may be new, ready, running, waiting or terminated state.
- **Program counter:** it indicates the address of the next instruction to be executed for this purpose.
- **CPU scheduling information:** This information includes process priority pointers to scheduling queues & any other scheduling parameters



Medium - Term Scheduler

- **CPU registers:** The registers vary in number & type depending on the computer architecture. It includes accumulators, index registers, stack pointer & general purpose registers, plus any condition- code information must be saved when an interrupt occurs to allow the process to be continued correctly after- ward
- **Memory management information:** This information may include such information as the value of the base & limit registers, the page tables or the segment tables, depending upon the memory system used by the operating system.
- **Accounting information:** This information includes the amount of CPU and real time used, time limits, account number, job or process numbers and so on.
- **I/O Status Information:** This information includes the list of I/O devices allocated to this process, a list of open files and so on. The PCB simply serves as the repository for any information that may vary from process to process.

CPU Scheduling Algorithm

CPU Scheduling deals with the problem of deciding which of the processes in the ready queue is to be allocated first to the CPU. There are four types of CPU scheduling that exist.

1. First Come, First Served Scheduling (FCFS) Algorithm
2. Shortest Job First Scheduling (SJF) Algorithm
3. Priority Scheduling Algorithm
4. Round Robin Scheduling Algorithm